

Stacking-order-driven interfacial dipole flip and chlorine-iodine exchange for high-performance perovskite-organic tandem solar cells

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Title

Stacking-Order-Driven Interfacial Dipole Flip and Chlorine-Iodine Exchange for High-Performance Perovskite-Organic Tandem Solar Cells

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Abstract

The physical adsorption of a self-assembled monolayer (SAM) onto an anchored layer introduces disordered stacking and a tangled dipole orientation at the hole-selective layer (HSL) surface, which affects the charge-transport dynamics and interface-driven energy loss in photovoltaic devices. Herein, we develop a collaborative strategy that combines stacking-order-driven interfacial dipole flipping with a chlorine-iodine exchange, achieved by depositing bismuth oxychloride (BiOCl) onto SAMs. This HSL exhibits a crystalline order characterized by dipole flipping and long-range periodic alignment in in-plane direction, driven by π - π interactions among carbazole units. It demonstrates enhanced stability against perovskite precursor solvents and thermal perturbations. The chlorine from BiOCl undergoes a chlorine-iodine exchange at the buried interface, modulating perovskite crystallization and weakening electron-phonon coupling. Our champion single-junction inverted perovskite solar cells (PSCs) achieve remarkable power conversion efficiencies (PCEs) of 27.25% (1.55 eV, 0.09 cm²) and 20.24% (1.85 eV, 0.09 cm²) with exceptional repeatability and operational stability. When employing SAMs/BiOCl as both HSL and interconnection layer, the champion perovskite-organic tandem solar cell (PO-TSC) attains an exceptional PCE of 28.14% (certified 27.84%) with steady power output of 27.86% (certified 27.59%). The device retains 90% of its initial efficiency under continuous maximum power point tracking with one-simulated-sun illumination over 600 hours.

Keywords: Interfacial dipole flip; Chlorine-iodine exchange; Bismuth oxychloride; Self-assembled monolayer; Perovskite-organic tandem solar cells

Introduction

Tandem solar cells (TSCs) have garnered significant attention due to their potential to surpass the intrinsic Shockley-Queisser efficiency limit of single-junction photovoltaic devices.¹⁻³ Compared to perovskite-inorganic TSCs (e.g., perovskite-copper indium gallium selenide (CIGS), perovskite-silicon and perovskite-perovskite configurations), perovskite-organic TSCs are composed of a wide-bandgap perovskite front cell and an absorption-complementary narrow-bandgap organic bulk-heterojunction rear cell, which offer distinct advantages (e.g., solution processability and exceptional flexibility). These attributes enable their application in diverse scenarios, including building/vehicle-integrated photovoltaics, park/street fencing, consumer electronics, aerospace vehicles and space stations.⁴⁻⁶ The development of near-infrared-adsorbing organic materials has been intensively pursued to extend the light-adsorption range of the rear cell.^{2,7} Meanwhile, numerous studies have focused on mitigating halogen separation and stabilizing the perovskite phase in the front cell.^{1,8-12} However, the certified power conversion efficiency (PCE) of 26.7%² for reported perovskite-organic TSCs remains lower than that of single-junction perovskite solar cells (27.2%)¹³. This discrepancy is primarily attributed to the significant challenges in simultaneously minimizing charge-transport losses at the interconnecting layer and the buried interfaces of the light-adsorbed layer.¹⁴⁻¹⁷

By systematically evaluating the reported interconnection engineering approaches (e.g. $\text{C}_{60}/\text{SnO}_{2-x}$,¹⁸ $\text{MoO}_3/\text{SnO}_x$,¹⁹ $\text{MoO}_3/\text{Au}/\text{ZnO}$,²⁰ $\text{C}_{60}/\text{YbO}_x/\text{Au}/\text{MoO}_x$,²¹ $\text{MoO}_3/\text{PEDOT:F}$,²² and MoO_3/SAMs ^{9,14}) alongside interface and photo-adsorbed layer engineering strategies,^{4,5} we demonstrate that fabricating a metal oxide-SAM hybrid hole-transporting layer can significantly

reduce production costs and enhance batch stability in tandem devices. This approach eliminates the need for high-cost atomic layer deposition techniques. On the other hand, this approach has the potential to mitigate charge-transport losses by elevating the work function (WF) of the metal oxide and inducing band bending in the photo-adsorbed layer.^{9,14} The weak anchoring interaction of phosphonic acid with the metal oxide surface results in a portion of the self-assembled monolayers (SAMs) being anchored to the surface, forming a well-defined anchored layer. However, the majority of SAMs engage in intermolecular hydrogen bonding or van der Waals interactions, leading to the formation of physically adsorbed self-aggregates characterized by inhomogeneous distribution, disordered stacking, and tangled dipole orientations.^{23,24} These issues detrimentally impact charge transport and extraction, thereby exacerbating interface-driven energy losses. To date, several strategies have been reported to address the challenges associated with physically adsorbed self-assembled monolayers (SAMs). For instance, developing novel SAMs with an extended π -skeleton on the carbazole unit can enhance intermolecular π - π stacking and achieve a uniform interface contact.²⁵⁻²⁷ Additionally, additive engineering approaches, such as constructing co-adsorbed layers,²⁸ co-assembly layers,²⁹ layer-by-layer assembly bilayers,³⁰ and hybrid assembly layers³¹, have been employed to mitigate the self-aggregation of SAMs on metal oxide surfaces. However, the issues of stacking order and dipole direction in physically adsorbed SAMs remain largely unresolved.

Herein, we develop an effective collaborative strategy that combines stacking-order-driven interfacial dipole flipping with chlorine-iodine exchange, achieved by introducing bismuth oxychloride (BiOCl) onto the surface of self-assembled monolayers (SAMs). This approach addresses the long-standing challenges associated with physically adsorbed SAMs. The strong interfacial bonding interaction between the alternating stacking-order $[\text{Bi}_2\text{O}_2]^{2+}$ and phosphonic

acid results in the adsorbed SAMs exhibiting a crystalline order characterized by dipole flipping and long-range periodic alignment in the in-plane (IP) direction, facilitated by π - π interactions among carbazole units. This hierarchical structure (HSL) demonstrates significantly enhanced stability against perovskite precursor solvents and thermal perturbations. Additionally, the chlorine from BiOCl undergoes a chlorine-iodine exchange at the buried interface, effectively modulating the crystallization of the upper perovskite layer and successfully converting tensile stress into compressive stress. This stress transformation weakens electron-phonon coupling within the perovskite, thereby enhancing its stability. By leveraging the synergistic interaction between stacking-order-driven interfacial dipole flipping and chlorine-iodine exchange, our champion single-junction inverted photovoltaic devices achieve remarkable power conversion efficiencies (PCEs) of 27.25% (1.55 eV bandgap, 0.09 cm²), 25.88% (1.55 eV bandgap, 1.0 cm²), and 20.24% (1.85 eV bandgap, 0.09 cm²), respectively, with exceptional repeatability and operational stability. Furthermore, when employing the SAMs/BiOCl heterostructure as both the hole-transporting layer (HSL) and interconnection, our champion perovskite-organic tandem solar cell (PO-TSC) delivers an exceptional PCE of 28.14% (certified 27.84%) with a steady power output of 27.86% (certified 27.59%). The device exhibits T90 operational stability (retaining 90% of its initial efficiency) over 600 hours under continuous maximum power point (MPP) tracking with one-simulated-sun illumination (AM 1.5AG, 100 mW cm⁻²).

Results

The synthesis of BiOCl, working mechanism of BiOCl on adsorbed Me-4PACz and its influence on HSL

Bismuth oxychloride (BiOCl), a nano-layered crystal structure characterized by alternating [Bi₂O₂]²⁺ slabs and halide layers with an average thickness of 3.5 nm, was synthesized and

characterized by high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) and X-ray diffraction (XRD) (Supplementary Fig. 1-3). The BiOCl obtained in this study exhibited a well-defined layering and stacking-order structure, along with chemically active surfaces and inherent polarity at the nanoscale. These properties were anticipated to make a significant contribution to the SAMs/perovskite interface. More importantly, the obtained bismuth oxychloride (BiOCl) exhibited excellent dispersion stability in ethyl alcohol solution. The freshly prepared BiOCl nanosheet exhibited a narrow and unimodal size distribution with an average diameter of 120 nm. After storage for 200 h, the average particle size slightly increased to 131 nm (Supplementary Fig. 4 and 5). Ultraviolet visible diffuse reflectance spectroscopy (UV-vis DRS) reflectance spectroscopy was performed to determine the optical bandgap of BiOCl nanosheets, yielding a bandgap of 3.64 eV (Supplementary Fig. 6). The ultraviolet photoelectron spectroscopy (UPS) results showed that the valence band top (VBM) was located at -7.88 eV (relative to the vacuum level), and the Fermi level (E_F) was located at -6.14 eV (Supplementary Fig. 7 and Table 2). Combined the bandgap value of 3.64 eV, the conduction band bottom (CBM) position was calculated to be -4.24 eV. Compared with the difference between the Fermi level and the conduction band bottom (1.90 eV), the Fermi level is closer to the valence band top with the difference of 1.74 eV, indicating that holes are the main carriers. Building on this foundation, we initially investigated the interfacial interactions between BiOCl and 4-(3,6-dimethyl-9H-carbazol-9-yl)butyl] phosphonic acid (Me-4PACz) using density functional theory (DFT) calculations, with Me-4PACz serving as a hole-selective layer. Me-4PACz was adsorbed strongly onto the BiOCl surface, with an adsorption energy of -7.31 eV (Fig. 1a). Furthermore, the electrons surrounding the phosphonic acid in Me-4PACz and the BiOCl surface were redistributed, indicating a strong interfacial affinity of the phosphonic acid for the $[\text{Bi}_2\text{O}_2]^{2+}$ layer. These interfacial interactions

were further corroborated by the shifts in P 2*p* signals from P–OH and P=O groups to lower binding energies, along with the movement of O 1*s* signal from the Bi–O unit to a higher binding energy (Supplementary Fig. 8). The opposite movements align well with the shift of the Bi–O unit from 1081.69 to 1116.05 cm⁻¹ in the Fourier transform infrared spectroscopy (Supplementary Fig. 9). Additionally, the observed shifts in the Cl 2*p* (from 199.44 and 197.83 eV to 199.58 and 197.97 eV) and Bi 4*f* (from 164.65 and 159.32 eV to 164.45 and 159.12 eV) signals further confirmed the strong interfacial interactions between BiOCl and Me-4PACz (Supplementary Fig. 10).

The molecular packing and orientation of Me-4PACz upon BiOCl treatment were elucidated using grazing-incidence wide-angle X-ray scattering (GIWAXS). For the Me-4PACz, no scattering rings were observed in the horizontal or vertical directions (Supplementary Fig.11), indicating a short-range disordered amorphous crystallinity. However, we observed a scattering ring extending from the horizontal direction (at $q_{xy} = 0.95 \text{ \AA}^{-1}$), suggesting an enhanced crystalline order and long-range periodic alignment in the in-plane (IP) direction. This phenomenon was primarily attributed to the π - π stacking among carbazole units in the adsorbed Me-4PACz molecules, driven by the interfacial bonding interaction with the alternating [Bi₂O₂]²⁺ structure. In situ GIWAXS was conducted to real-time monitor the dynamic changes in molecular packing and orientation of Me-4PACz (Fig.1b and 1c). At the early stage of the annealing process (before 20 s), Me-4PACz exhibited a clear out-of-plane (OOP) orientation, with the diffraction peak centered at 0.52 Å⁻¹ (Fig.1b and 1c).

Subsequently, the out-of-plane (OOP) diffraction peak vanished and was replaced by an in-plane (IP)-dominant diffraction peak, further demonstrating that bismuth oxychloride (BiOCl) effectively regulated the ordered-arrangement orientation evolution of Me-4PACz molecules. Additionally, the overall diffraction signal intensity of Me-4PACz films was significantly

enhanced, and a novel π - π stacking diffraction peak emerged at $0.95 \text{ \AA}^{-1}$ upon BiOCl modification (Supplementary Fig. 12). To exclude the interference from BiOCl diffraction ring, the optimized stacking configuration of the SAMs was further conducted by XPS characterization. As shown in Supplementary Fig. 13, there is a clearly peak at 291.2 eV in C 1s spectra, which indicates a strong π - π stacking within the SAMs. Both observations confirmed that the strong interfacial bonding interaction between the phosphonic acid and the $[\text{Bi}_2\text{O}_2]^{2+}$ structure promoted the ordered arrangement of adsorbed Me-4PACz molecules, characterized by enhanced crystallinity and dipole flipping. To elucidate these findings, a schematic model was proposed to illustrate the interfacial dipole flipping and ordered stacking of adsorbed Me-4PACz. As shown in Fig. 1d, the adsorbed Me-4PACz initially exhibited a disordered packing on the anchored layer. Upon BiOCl treatment, the adsorbed and disordered Me-4PACz transformed into a long-ordered arrangement with π - π interactions and dipole flipping, compared to the anchored Me-4PACz layer. This phenomenon was designated as "stacking-order-driven interfacial dipole flip" and resulted in a more robust hole-selective layer (HSL).

To verify the stability of the hole-selective layer (HSL), we subjected it to separate treatments of N,N-dimethylformamide (DMF) rinsing and thermal aging, followed by an examination of the changes in its surface contact potential difference (CPD) using Kelvin probe force microscopy (KPFM). As shown in Fig. 1e and 1f, the surface potential of pristine Me-4PACz exhibited a significant average CPD shift ($\Delta\text{CPD} \approx 270 \text{ mV}$) after DMF rinsing, along with substantial broadening of the CPD distribution. These changes persisted after continuous heating at 85°C for 200 hours, indicating that pristine Me-4PACz was prone to desorption and thermally unstable. In contrast, after BiOCl modification, the average CPD shift was reduced to 34 mV following DMF rinsing, and the CPD distribution remained nearly unchanged after 200 hours of heating at 85°C .

(Fig. 1g and h). These results demonstrated a highly stable and uniformly distributed HSL. Conductive force microscopy (CFM) measurements revealed that, compared to pristine Me-4PACz (Fig. 1i and j), the BiOCl-modified Me-4PACz exhibited a more uniform current distribution with minimal degradation after DMF rinsing and thermal aging (Fig. 1k and 1l). This demonstrated significantly enhanced stability against perovskite precursor solvents and thermal perturbations.³⁴ This stability is advantageous for improving the conductivity and carrier mobility of the hole-selective layer (HSL). As shown in Supplementary Fig. 14, BiOCl modification increases the conductivity from $1.96 \times 10^{-4} \text{ S cm}^{-1}$ to $2.57 \times 10^{-4} \text{ S cm}^{-1}$, and increases the hole mobility from 3.66×10^{-3} to $4.95 \times 10^{-3} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$. Then, Hall effect measurement of original SAM and BiOCl-modified SAM shows that the hole mobility increases from $20.2 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ to $26.2 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ after BiOCl modification (Supplementary Fig. 15).

The influence of BiOCl on perovskite crystallization and the working mechanism

To investigate the influence of BiOCl modification on perovskite crystallization and growth, we spin-coated perovskite precursor solutions onto both pristine and BiOCl-modified Me-4PACz substrates. In situ photoluminescence (PL) microscopy images revealed that the perovskite precursor exhibited faster nucleation but slower crystallization growth compared to the control case (Supplementary Fig. 16), which was conducive to achieving high-quality perovskite films.³⁵ To elucidate this process in detail, in situ PL measurements also showed that the perovskite precursor underwent immediate nucleation but required a longer time (28 s) to reach full crystallization than the control (16 s) during thermal annealing (Supplementary Fig. 17a and 17b). These processes were further corroborated by in situ ultraviolet-visible (UV-vis) spectroscopy, which revealed that the perovskite precursor required a longer time (25 s) than the control (18 s) to achieve strong adsorption intensity (Supplementary Fig. 17c and 17d). The above in situ

observations demonstrated that bismuth oxychloride (BiOCl) effectively modulated perovskite crystallization dynamics by accelerating the nucleation of the precursor solvent and delaying the crystal-grain growth.³⁵

The crystallization quality of the obtained perovskite film was initially evaluated using X-ray diffraction (XRD). As shown in Supplementary Fig. 18, the intensities and the ratio of the (100) and (110) diffraction peaks were significantly enhanced, indicating improved crystallinity and crystal orientation. This finding was further corroborated by grazing-incidence wide-angle X-ray scattering (GIWAXS) measurements of the perovskite top surface (Fig. 2a and 2b), where the BiOCl-modified perovskite exhibited preferential alignment of the (100) planes along the 45° and 135° azimuthal angles. This orientational crystallization was consistent with the buried surface GIWAXS images (Supplementary Fig. 19). Additionally, the top surface of the perovskite grown on the Me-4PACz/BiOCl substrate displayed a smoother and more uniform morphology, with a root-mean-square (RMS) roughness of 22.18 nm and a larger average crystallographic size of 650 nm, compared to the control sample (RMS = 32.35 nm, average crystallographic size = 480 nm, Supplementary Fig. 20 and 21). As further confirmed by time-resolved confocal photoluminescence (PL) intensity and lifetime mapping (Supplementary Fig. 22 and 23), the perovskite grown on the Me-4PACz/BiOCl substrate exhibited higher overall PL intensity and a longer PL lifetime (13340–53906 ps) compared to the control (605–15783 ps). These improvements in perovskite quality were primarily attributed to the enhanced surface wettability (Supplementary Fig. 24) and reduced surface roughness (Supplementary Fig. 25) of the Me-4PACz/BiOCl substrate. These surface modifications facilitated a more uniform spreading of the perovskite precursor and provided greater opportunities for the Cl⁻ layer in BiOCl to regulate the crystallization of the perovskite precursor. As supported by the bottom surface morphology of the

perovskite (Supplementary Fig. 26 and 27), the perovskite precursor spin-coated on the Me-4PACz/BiOCl substrate demonstrated a significant reduction in PbI₂ residual and an enhanced average crystallographic size.

To elucidate the underlying mechanisms responsible for the aforementioned improvements in crystallization and surface morphology, high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) was employed. As shown in Fig. 2c and 2d, the cross-sectional BiOCl/perovskite interface exhibited a tight atomic-scale contact, wherein Cl⁻ ions interacted closely with I⁻ ions from the perovskite, leading to a Cl-I exchange process. These interactions resulted in a highly ordered and stable interface that effectively modulated the crystallization process of the upper perovskite layer.¹³ The Cl-I exchanges were further confirmed as a thermodynamically favorable process with a minimal energy change ($\Delta E = 0.016$ eV) through density functional theory (DFT) calculations (Fig. 2e). To further verify the interfacial interactions, X-ray photoelectron spectroscopy (XPS) was performed. As shown in Fig. 2f and Supplementary Fig. 28, a group of novel Pb 4f peaks (141.91 and 137.20 eV) attributed to Pb-Cl bonds, along with I 3d peaks (629.04 and 617.70 eV) belonging to the BiOI structure, were observed from the BiOCl/PbI₂ film.³³ These findings effectively confirm that a Cl-I exchange occurred at the BiOCl/perovskite interface. Furthermore, Fourier transform infrared spectroscopy (FTIR) revealed noticeable shifts in the Bi-O and Bi-Cl peaks for the BiOCl/PbI₂ film (Supplementary Fig. 29). Additionally, time-of-flight secondary ion mass spectrometry (TOF-SIMS) demonstrated that Cl⁻ signals were distributed at both the BiOCl/perovskite interface and the buried surface of the perovskite layer (Fig. 2g). There is almost no chlorine (Cl) signal observed in control perovskite bulk phase, demonstrating that the Cl signal detected in the perovskite bulk phase originates from the spontaneous Cl-I ion exchange between BiOCl and perovskite rather than from the MACl

additive in the original precursor (Supplementary Fig. 30). These results provided direct evidence of the Cl-I exchange at the interface and the infiltration of Cl^- into the perovskite crystal, thereby modulating the crystallization of the upper perovskite layer.

The influence of the Cl-I exchange on the residual stress within the perovskite was investigated using grazing incidence X-ray diffraction (GIXRD). As depicted in Fig. 2h and Supplementary Fig. 31, the tensile stress in the perovskite (28.81 MPa) was gradually transformed into compressive stress (-21.28 MPa) following the modification of BiOCl, according to Supplementary Note 1. This high compressive stress was beneficial for mitigating electron-phonon coupling. Temperature-dependent photoluminescence (PL) measurements (Fig. 2i and Supplementary Fig. 32) demonstrated that as the temperature increased from 93 to 293 K, the PL intensity progressively rose, reaching its peak at 233 K with broadened emission signals, indicating a reduction in lattice vibrations. By fitting the relationship between temperature and the full width at half-maximum (FWHM) of the photoluminescence (PL) signals, by using Supplementary Note 2, the values of Γ_0 and γ_0 for the BiOCl-modified perovskite were determined to be 33.26 meV and 19.85 meV, respectively. These values were lower than those for the control sample ($\Gamma_0 = 35.61$ meV and $\gamma_0 = 27.93$ meV), indicating reduced carrier scattering and suppressed electron-phonon coupling.³⁶ These effects contributed to enhanced perovskite stability and improved carrier transport.

The stability of the perovskite films was evaluated by exposing them to an air atmosphere with a relative humidity (RH) of $30 \pm 5\%$ at 25°C . The duration for maintaining the competitive black phase of the perovskite was extended to over 60 days (Supplementary Fig. 33) following BiOCl modification, as further confirmed by the corresponding atomic force microscopy (AFM) images (Supplementary Fig. 34). This enhanced stability could be attributed to the high crystallinity,

hydrophobic perovskite surface (Supplementary Fig. 35), reinforced interface, and superior thermal stability, which exhibited no attenuation in the adsorption signal after 2 hours of continuous heating at 85°C in air (Supplementary Fig. 36). Additionally, upon BiOCl modification, the photoluminescence (PL) response time was extended to over 120 minutes under continuous UV light exposure in air (Supplementary Fig. 37), indicating superior photostability.

The influence of BiOCl on charge-transporting dynamics

The carrier transport was initially investigated by steady-state photoluminescence (PL) and time-resolved PL (TRPL), respectively. The PL intensity took a significant reduction for the perovskite grown onto Me-4PACz/BiOCl substrate when illuminating from glass-side (Supplementary Fig. 38). The average carrier decay lifetime was reduced from 8267.75 to 1511.74 ns (Supplementary Note3, Fig. 39, and Table 1), supporting the effective interface charge transport. As further confirmed by the transient absorption (TA) spectroscopy in Supplementary Fig. 40, a faster decay of the ground-state bleach peak was observed. The corresponding and fitting kinetic curves revealed that the perovskite its saturation with slower decay and weaker intensity after BiOCl modification, confirming a reduced interfacial non-radiative recombination and an enhanced carrier extraction.³¹ To explore the deep reason for above enhanced charge transport dynamics, we simulated the charge transfer between the (110) facet of BiOCl and the (100) facet of perovskite by DFT. As shown in Supplementary Fig. 41 and 42, under optimal conditions, the plane-averaged charge distributed at the BiOCl/perovskite, in which the charge transfer of 2.21 e⁻ occurred from perovskite to BiOCl. Moreover, when BiOCl interacts with Me-4PACz, charge transfer occurred from BiOCl to Me-4PACz. These simulated results indicated that BiOCl facilitated hole extraction and transport at the buried interface. The interfacial energy alignment was further clarified by UPS measurements. As shown in Supplementary Fig. 43-46 and Table 2, the BiOCl modification shifts

the valence-band maximum of the HSL from -5.18 eV for Me-4PACz to -5.51 eV for Me-4PACz/BiOCl, while the valence-band maximum of the perovskite is located at -5.69 eV. This corrected energy-level alignment is favorable for hole extraction from the perovskite to the HSL and supports the reduced interfacial charge-transport barrier at the buried interface.³¹ As further supported by a lower reverse saturation current in region I and a steeper slope in region II (Supplemental Fig. 47) as well as a faster photocurrent decay (4.71 μ s) and a slower photovoltage decay (34.97 μ s) (Supplementary Fig. 48) for the BiOCl-modified device. These results greatly attributed the reduced disorder in the energy levels within perovskite and the reduced interfacial defects density, confirmed in Supplementary Fig. 49 and 50, respectively.

The device performance of the inverted single-junction perovskite solar cells based on Me-4PACz/BiOCl substrate

To evaluate the influence of Me-4PACz/BiOCl substrate on photovoltaic performance, an inverted device with the structure of FTO/Me-4PACz/BiOCl/perovskite/ C_{60} /BCP/Ag was fabricated with 1.55 eV FAMACsPbI₃ as active layer (Fig. 3a). The optimal concentration of BiOCl was determined to be 1.0 mg mL⁻¹ (Supplementary Fig. 51 and Table 3). Current-voltage (J - V) curves showed that the BiOCl-modified device (0.09 cm²) achieved a PCE of 27.25% with an increased open-circuit voltage (V_{OC}) of 1.200 V, an improved fill factor (FF) of 86.98%, a steady output PCE of 27.11% (Fig. 3b), and a negligible hysteresis index ($HI = 2.42\%$, Supplementary Fig. 52 and Table 4). The integrated current density from external quantum efficiency (EQE) curve was determined as 25.79 mA cm⁻² (Supplementary Fig. 53), closely matching the 26.11 mA cm⁻² obtained from the J - V curves. The repeatabilities of these devices are assessed through statistical analysis of forty independent devices (Supplementary Fig. 54 and 55), in which the BiOCl-modified devices exhibited an average PCE of 26.98%, along with the enhanced V_{OC} and FF .

With the incorporation of BiOCl, the built-in potential (V_{bi}) increased from 1.11 to 1.17 V (Supplementary Note 5 and Fig. 56) and the V_{OC} loss induced by non-radiative recombination decreased from 93 to 71 mV (Fig. 3c), both of which contributed to an increase in V_{OC} . Moreover, the photoluminescence quantum yield (PLQY) value was enhanced from 0.06 for control to 0.14 for BiOCl-modified device (Supplementary Fig. 57). This higher PLQY resulted in an elevation of the quasi-Fermi level splitting (QFLS) according to Supplementary Note 6, further reducing carrier transport barriers, suppressing non-radiative recombination at the HSL/perovskite interface, and thus enhancing V_{OC} (Supplementary Note 7).³⁷ To assess the effect of BiOCl modification on fill factor (FF) loss, the light intensity-dependent V_{OC} relationship was investigated. As shown in Fig. 3d, the BiOCl-modified device had a smaller ideality factor ($1.23 K_B T/q$) than the control device ($1.67 K_B T/q$). According to Supplementary Note 8 and 9, this smaller ideality factor indicated an enhancement in charge-carrier transport and a reduction in non-radiative recombination, thereby contributing to a decrease in FF loss (Supplementary Fig. 58). Additionally, a lower series resistance ($R_s = 2.97 \Omega$) and a higher recombination resistance ($R_{rec} = 3130 \Omega$) observed from electrochemical impedance spectroscopy (Supplementary Fig. 59 and Table 5) illustrated that incorporating BiOCl at the HSL/perovskite interface significantly enhanced charge-carrier transport and alleviated non-radiative recombination, leading to reduced V_{OC} and FF losses.³⁸ Thanking to the above reasons that enhanced charge transfer and reduced non-radiative recombination, the 1.0 cm^2 active-area inverted device achieved a PCE of 25.88% (Fig. 3e and Supplementary Table 6). Further, we replaced Me-4PACz with 2-(3,6-dimethoxycarbazol-9-yl)ethylphosphonic acid (MeO-2PACz) and 4-(7H-dibenzo[c,g]carbazol-7-yl)butylphosphonic acid (4PADCB), respectively, to fabricate the inverted devices. As shown in Fig. 3f and Supplementary Table 7, after the modification of BiOCl, the photovoltaic performances of these

devices were significantly enhanced, demonstrating the compatibility of BiOCl modification on different SAM-based devices.

The operational stability of these devices was evaluated under continuous when heating at 85°C in nitrogen-filled glove box. After 1200 hours of aging, the BiOCl-modified device exhibited an initial efficiency retention of 91.92%, while the control device only retained 60.76% (Supplementary Fig. 60). Moreover, when the BiOCl-modified device operated under a constant reverse bias voltage of $-V_{OC} = -1.20$ V according to the ISOS-V-1 protocol, it showed superior reverse bias resilience, retaining 85.13% of its initial efficiency after 1000 hours (Supplementary Fig. 61). When these devices were subjected to continuous illumination (1-sun, AM1.5G, 100 mW cm⁻²) at the maximum power point (MPP), the BiOCl-modified device maintained 95.24% of its initial PCE over 2400 hours, which was significantly higher than 52.13% retained by the control device (Fig. 3g). These results clearly demonstrated that inserting BiOCl at Me-4PACz/perovskite interface significantly enhanced the operational stability of the inverted device, attributing to the enhanced stability of HSL and the perovskite layer.

To verify the generalizability of the Me-4PACz/BiOCl substrate for other bandgap light-adsorption layer, we utilized 1.85 eV wide-bandgap (WBG) perovskite to fabricate inverted PSCs. As shown in Supplementary Fig. 62, the BiOCl-modified device attained a PCE of 20.24% (the highest PCE of 1.85 eV bandgap single-junction PSCs, Supplementary Table 8) with an FF of 87.74%, a V_{OC} of 1.389 V, a J_{SC} of 16.61 mA cm⁻², and a well repeatability (Supplementary Fig. 63 and 64). The obtained J_{SC} value was well agreement with the integrated current (16.65 mA cm⁻²) from EQE curves (Supplementary Fig. 65). The relatively high J_{SC} is partly associated with enhanced short-wavelength absorption of the perovskite films (Supplementary Fig. 66a and b), contributing to the increased photocurrent. This exceptional photovoltaic performance was

ascribed to the reduced FF and V_{oc} losses (Supplementary Fig. 67). More importantly, this device exhibited a steady-output PCE of 19.42% (Supplementary Fig. 68) with a low HI of 1.08% (Supplementary Fig. 69 and Table 9), indicating that the ionic migration in wide bandgap perovskite was effectively restricted because of the residual stress conversion from tensile to compressive.

The device performance of the perovskite-organic tandem solar cells

Considering the excellent generalizability of Me-4PACz/BiOCl substrate, we anticipated it could replace the traditional interconnections in perovskite-organic tandem solar cells (PO-TSCs), such as C_{60}/SnO_{2-x} , MoO_3/SnO_x , $MoO_3/Au/ZnO$, and $C_{60}/YbO_x/Au/MoO_x$.¹⁸⁻²² As is all-known, fabricating these traditional interconnections required the high-cost atomic layer deposition technique, which increased the batch instability and the device cost. More importantly, the surface work function of SnO_x or other metal oxide could not be effectively regulated.^{4,5} Before constructing the PO-TSCs, we evaluated single-junction OSCs based on different organic active-layer systems. As shown in Supplementary Fig. 70 and Table 10 and 11, the BiOCl-modified HSL improves the photovoltaic performance of PM6: BTP-eC9 and PM6: BTPSe-Ph4F based OSCs, respectively, supporting the compatibility of this strategy with different OSC systems. Based on the above accounts, we fabricated the PO-TSC with the structure of FTO/Me-4PACz/BiOCl/1.85 eV perovskite/ $C_{60}/BCP/Au/MoO_x$ /Me-4PACz/BiOCl/OSC/PNDIT-F3N/Ag (Fig. 4a), in which PM6 and BTP-eC9 were selected to prepare the organic bulk heterojunction (BHJ) layer and PNDIT-F3N was selected as cathode interface layer (Supplementary Fig. 71). The thicknesses of the perovskite and organic BHJ layers were optimized to approximately 380 and 120 nm (Fig. 4b), respectively, ensuring efficient photon absorption and current matching between the two subcells. By utilizing Me-4PACz/BiOCl as HSL, the PO-TSC achieved a PCE of 28.14% with a V_{oc} of

2.212 V, a J_{SC} of 15.05 mA cm⁻², an FF of 84.53% (Fig. 4c), a steady output efficiency of 27.86% (Fig. 4d), and a HI of 3.00% (Supplementary Fig. 72 and Table 12). The slight enhancement of J_{SC} is on the one side attributed the higher transmittance of MgF₂-coated FTO than that of bare FTO, particularly in the short-wavelength region (Supplementary Fig. 73). On the other side, BiOCl modification improves the wettability of the interconnecting layer for subsequent organic active-layer deposition. As proved in Supplementary Fig. 74, the water contact angle of the FTO/HSL/perovskite/ETL/ICL decreases from 74.8° to 52.9° after BiOCl modification, which facilitates the spreading of the PM6 precursor solution and leading to a more ideal bi-continuous interpenetrating network and higher crystallinity in the BTP-eC9 film (Supplementary Fig. 75). The reproducibility of the PO-TSCs was further verified by statistical analysis of 40 independent devices, where the BiOCl-modified devices show consistently enhanced FF, V_{OC} and PCE compared with the control devices (Supplementary Fig. 76 and 77). Additionally, we sent these devices to the Chengdu Institute of Product Quality Inspection Co., Ltd. National Photovoltaic Product Quality Inspection & Testing Center, and obtained a certified PCE of 27.84% and a steady output PCE of 27.59% (Supplementary Fig. 78). These values represent the highest reported and certified efficiency for PO-TSCs to date (Fig. 4e and Supplementary Table 13). Additionally, according to the EQE spectra (Fig. 4f), the J_{SC} values integrated from the EQE spectra for the front and rear cells were 15.06 and 14.91 mA cm⁻², respectively, demonstrating perfect current matching with the J_{SC} values from the J - V measurements. We have also applied the BiOCl-based interface to prepare PO-TSCs with 1.81 and 1.83 eV wide bandgap perovskites as front cells, respectively, yielding PCE of around 27.50% (Supplementary Fig. 79 and Table 14). The stability of the PO-TSCs was also evaluated through continuous MPP tracking under one simulated-sun illumination (AM1.5AG, 100 mW cm⁻²), from which the PO-TSCs retained T_{90} (the time to retain 90% of the

initial efficiency) after 600 hours (Fig. 4g).

Discussion

We demonstrate that the introduction of stacking-ordered BiOCl onto SAMs results in the adsorbed SAMs exhibiting a crystalline order characterized by dipole flipping and long-range periodic alignment in the in-plane (IP) direction. This phenomenon is driven by π - π interactions among carbazole units and a chlorine-iodine exchange occurring at the buried interface. Consequently, this mechanism enhances the stability of the HSL and modulates the crystallization of the upper perovskite layer. Owing to these contributions, our champion single-junction inverted PSCs achieve remarkable PCEs of 27.25% (1.55 eV, 0.09 cm²) and 20.24% (1.85 eV, 0.09 cm²), respectively, with exceptional repeatability and operational stability. When utilizing the SAMs/BiOCl as both the HSL and interconnection layer, our champion PO-TSC outputs an exceptional PCE of 28.14% (certified 27.84%) with a steady power output of 27.86% (certified 27.59%) and excellent MPP operational stability. We anticipate that this collaborative strategy, combining stacking-order-driven interfacial dipole flipping and chlorine-iodine exchange, will significantly contribute to the development of stable perovskite/organic TSCs.

Methods

Materials

Formamidinium iodide (FAI), methylammonium chloride (MACl), methylammonium iodide (MAI), cesium iodide (CsI) and lead iodide (PbI₂) were obtained from Xi'an Yuri Solar Co., Ltd. N,N-dimethylformamide (DMF), dimethylsulfoxide (DMSO), 1,8-diiodooctane, chloroform (CF), chlorobenzene (CB), mannitol and anisole were purchased from Sigma Aldrich. Isopropanol (IPA) was purchased from adamas-beta. Fluorine-doped tin oxide (FTO) glass, nickel oxide (NiO_x) nanoparticle, fullerene (C₆₀), [6,6]-phenyl C₆₁ butyric acid methyl ester (PC₆₁BM) and

bathocuproine (BCP) were obtained from Advanced Election Technology Co., Ltd. (4-(3,6-Dimethyl-9H-carbazole-9-yl) butyl) phosphonic acid (Me-4PACz) was purchased from TCI (Shanghai) Development Co., Ltd. Bismuth nitrate pentahydrate ($\text{Bi}(\text{NO}_3)_3 \cdot 5\text{H}_2\text{O}$), polyvinylpyrrolidone (PVP), sodium chloride (NaCl) and magnesium fluoride (MgF_2) were obtained from adamas-beta. Molybdenum oxide (MoO_x) was purchased from Shanghai Aladdin Biochemical Technology. PM6, BTP-eC9 and PNDIT-F3N were purchased from Solarmer Materials. Chloroform (CF, purity of 99.90%) was purchased from Konosience. All reagents were used directly without further purification. Different masses of nanosheets were dispersed in ethanol and ultrasonically stripped for 8 hours to obtain BiOCl nano-dispersions with concentrations from 0, 0.5, 1, 1.5 to 2.0 mg mL^{-1} , respectively.

Synthesis of BiOCl nanosheets

Firstly, 0.324 g of $\text{Bi}(\text{NO}_3)_3 \cdot 5\text{H}_2\text{O}$ and 0.267 g PVP were dissolved in 16.5 mL mannitol solution (0.1 M) together. The mixture was then vigorously stirred for 10 min to form a uniform dispersion. Then, 3 mL saturated sodium chloride solution was added dropwise into the above mixture solution, and stirring continuously for another 10 minutes. The resulted white suspension was transferred to a teflon-lined high-pressure reactor and subjected to a hydrothermal reaction at 160°C for 6 h. After the reaction, the product was collected by centrifugation, rinsed three times with deionized water to remove impurities, and finally dried in a vacuum oven at 80°C overnight to yield the desired BiOCl material.

Perovskite precursor preparation

For the 1.55 eV $\text{FA}_{0.90}\text{MA}_{0.05}\text{Cs}_{0.05}\text{PbI}_3$ perovskite precursor solution, 0.0835 mmol MAI, 0.0835 mmol CsI, 1.503 mmol FAI and 1.67 mmol PbI_2 were dissolved in 1.0 mL mixture of DMF: DMSO (V:V, 4:1), and 20 mol% MACl was added to the solution for better crystallization. The precursor

solution was stirred at 298 K for at least 3 h and then filtered with a polytetrafluoroethylene membrane filter (0.22 μm) before use.

For the 1.85 eV $\text{FA}_{0.67}\text{MA}_{0.27}\text{Cs}_{0.06}\text{PbI}_{1.5}\text{Br}_{1.5}$ perovskite precursor solution, 0.066 mmol CsI, 0.737 mmol FAI, 0.297 mmol MAI, 0.275 mmol PbI_2 and 0.825 mmol PbBr_2 were dissolved into a 1.0 mL mixture of DMF: DMSO (V:V, 4:1). The precursor solution was stirred at 298 K for at least 3 h and then filtered with a polytetrafluoroethylene membrane filter (0.22 μm) before use.

For the preparation of the narrow organic precursor, PM6, BTP-eC9 and PC_{61}BM with a weight ratio of 1: 1.2: 0.3 were dissolved in CF, and the total concentration was optimized at 20 mg mL^{-1} . 0.5 vol% 1,8-diiodooctane was added to the solution, which was then stirred at 70 $^{\circ}\text{C}$ for 2 h before use.

Device fabrication

For normal bandgap perovskite, the FTO substrates were ultrasonically cleaned with glass cleaning solution, ultrapure water and ethanol, with each step lasting 20 min. The FTO substrates were treated with UV ozone for 30 min. The FTO substrates were then transferred immediately into a nitrogen-filled glove box and cooled down to the room temperature. 1.0 mg Me-4PACz was dissolved in 1.0 mL isopropanol to obtain SAM solution, which was spin-coated on FTO at 3000 rpm for 30 s and annealed at 100 $^{\circ}\text{C}$ for 10 min. The BiOCl nanosheet was subsequently spin-coated onto FTO/Me-4PACz substrate at 5000 rpm for 30 s and annealed at 100 $^{\circ}\text{C}$ for 10 min. The $\text{FA}_{0.90}\text{MA}_{0.05}\text{Cs}_{0.05}\text{PbI}_3$ perovskite (1.67 M) precursor solution was spin-coated onto the Me-4PACz film with or without BiOCl at 1000 rpm for 10 s and 5000 rpm for 30 s. At the last 5 s of second process, 150 μL chlorobenzene was instant and rapid dropped. The substrates were then transferred to the hot plate and annealed at 120 $^{\circ}\text{C}$ for 30 min. For the passivation treatment, 1 mg mL^{-1} PDAI_2 and 2 mg mL^{-1} 3MTPAI were added in CB: IPA = 1: 1 (V: V) separately, then mixed

and shook well in the ratio of 1:1 before use. The passivation treatment was processed on the perovskite film at 5000 rpm for 30 s, and then annealed at 100 °C for 10 min. The electron transfer layer (ETL) was obtained by evaporating 25 nm C₆₀ with a rate of 0.3 Å s⁻¹, and then 6 nm BCP at 0.3 Å s⁻¹ was evaporated as hole block layer. Finally, 100 nm silver electrode was evaporated at 1.0 Å s⁻¹ for obtaining the devices with 0.09 cm² and 1.00 cm², respectively. After device fabrication, a thickness of 110 nm MgF₂ was used as a reflective reduction layer on the back of FTO.

For 1.85 eV wide-bandgap perovskite, the preparation process was the same as above. The FA_{0.67}MA_{0.27}CS_{0.06}PbI_{1.5}Br_{1.5} perovskite precursor was spin-coated onto the Me-4PACz film with or without BiOCl at 1000 rpm for 10 s and 5000 rpm for 30 s. At the last 10 s of second process, 200 µL chlorobenzene was instant and rapid dropped, the substrates were then transferred to the hot plate and annealed at 100 °C for 10 min. The mixture of PDAI₂ and 3MTPAI was then spin-coated on the film at 5000 rpm for 30 s, and annealed at 100 °C for 10 min. 10 mg mL⁻¹ PC₆₁BM was prepared in CB solution, then spin-coated on above film at 2000 rpm. After that, 10 nm C₆₀ was evaporated with a rate of 0.3 Å s⁻¹, and 6 nm BCP at 0.3 Å s⁻¹ was then evaporated as hole block layer. Finally, 100 nm silver electrode was evaporated at 1.0 Å s⁻¹. After device fabrication, a thickness of 110 nm MgF₂ was used as a reflective reduction layer on the back of FTO.

For organic solar cell, the clean FTO was transferred into vacuum chamber for 15 nm MoOx evaporation. Then, Me-4PACz was spin-coated on MoOx as same as above process. The PM6:BTP-eC9: PC₇₁BM solution was dynamically cast onto the Me-4PACz with or without BiOCl layer at 3000 rpm for 30 s, then annealed at 100 °C for 5 min. For the ETL, 0.5 mg mL⁻¹ PNDIT-F3N with 0.5 vol% of acetic acid was dynamically spin-coated on the organic layer at 3000 rpm for 30s. Finally, 100 nm silver electrode was evaporated at 1.0 Å s⁻¹. After device fabrication, a thickness

of 110 nm MgF_2 was used as a reflective reduction layer on the back of FTO.

For monolithic PO-TSCs, the organic subcell was stacked on top of the perovskite subcell. After deposition of the BCP layer, a 1 nm Au interconnect was introduced by thermal evaporation. 15 nm MoOx was then evaporated under the same conditions described above, following the procedure used for the standalone organic subcells.

Characterization

Grazing-incidence wide-angle X-ray scattering (GIWAXS) measurements were conducted at the Shanghai Synchrotron Radiation Facility (SSRF) using a Huber five-circle diffractometer equipped with an EIGER X1M two-dimensional detector. The measurements were performed using a double-focused monochromatic X-ray beam at the corresponding beamline. For GIWAXS characterization, SAM and SAM/ BiOCl samples were prepared on cleaned Si substrates following the same deposition procedures used for the corresponding device substrates. The top surface and cross-sectional morphology of perovskite films were recorded on scanning electron microscope (SEM, thermo scientific Apreo 2C). Lattice characterization was completed on the transmission electron microscope (TEM) by using a Talos F200S G2 at an acceleration voltage of 200 kV and a current of less than 10 pA. The topography images of perovskite film were recorded by using atomic force microscope (AFM, Bruker NanoIR3). The water contact angle (CA) was tested by SDC-350. The crystallinity was measured via X-ray diffraction (XRD) on Bruker AXS D8 Advance with a Co-K X-ray irradiation source operated at 40 kV and 40 mA. Fourier transform infrared (FTIR) spectrometer was obtained from Vertex 70 infrared spectrometer (Bruker company, transmission mode, $\text{RES} = 2 \text{ cm}^{-1}$, scanning time = 16, scanning range = $400\text{--}4400 \text{ cm}^{-1}$). UV-vis absorption spectra were tested on the UV-3600i Plus. Photoluminescence (PL) and time-resolve PL (TRPL) spectra were measured by PicoQuant FluoTime 510-nm laser excitation. X-ray

photoelectron spectra were recorded on a Thermo Fisher Scientific ESCALAB 250Xi with a monochromatic Al K α X-ray source (1486.6 eV) and an X-ray beam of around 900 μ m (Czech Republic). Electrochemical impedance spectroscopy (EIS), Mott-Schottky (MS), transient photovoltage spectrum (TPV) and transient current spectrum (TPC) were measured on Paios. The J - V curves were collected on the reference silicon cell calibrated solar simulator, with Keithley model 2400 under 100 mW cm⁻² illumination and a simulated AM 1.5G spectrum. The EQE spectra were obtained on the Enlitech QE-R system.

Computational methods

The density functional theory (DFT) calculations were performed by using the Vienna Ab Initio Simulation Package (VASP). To describe the interactions between electrons and ions, as well as electron-electron interactions, the Projector Augmented-Wave (PAW) pseudopotential was employed, along with the Perdew-Burke-Ernzerhof (PBE) functional for exchange-correlation. The plane-wave cutoff energy was set to 520 eV, with convergence criteria for total energy and force set to 10⁻⁸ eV and 0.02 eV/Å, respectively. During the structural optimization of BiClO(110)/FAPbI₃(100), the Brillouin zone sampling for the KPOINTS was performed by using a 4×4×1 grid scheme. The three-dimensional visualization of crystal, volumetric and morphology data were conducted by using VESTA. The work functions were calculated by using the latest implementation of the Gaussian and Plane Waves (GPW) formalism in the Quickstep module of the CP2K program package, where the exchange-correlation term was treated by employing the Perdew–Burke–Ernzerhof (PBE) function. Grimme's DFT-D3 corrections was adopted to treat the Van der Waals interactions. Molecular orbitals of the valence electrons were expanded into DZVP-MOLOPT-SR-GTH basis sets, while atomic core electrons were described through Goedecker-Teter-Hutter pseudopotentials. For the auxiliary plane wave expansion of the charge density, an

energy cutoff of 450 Ry and real cutoff of 60 Ry were applied. Geometry optimizations were performed using the Broyden–Fletcher–Goldfarb–Shanno (BFGS) minimization algorithm. The structures were optimized until atomic displacements were less than 3×10^{-3} Bohr, and the forces were below 4.5×10^{-4} Hartree/Bohr.

The stability of devices

For air stability, after the same cleaning steps for FTO, 30 min UVO treatment, Me-4PACz, and BiOCl preparation were sequentially carried out in the same manner as described above for device preparation. FAMACsPbI₃ were prepared on NiOx/Me-4PACz and NiOx/Me-4PACz/BiOCl substrates, respectively. Subsequently, the prepared perovskite films were placed in air (RH = 30±5%, $T = 25^\circ\text{C}$) for a series of tests.

For thermal stability, the unencapsulated devices were positioned within a nitrogen-filled glove box and continuously maintained at 85 °C. Device performance was evaluated at regular intervals, and the power conversion efficiency (PCE) was meticulously recorded.

For maximum-power-point tracking, maximum power point (MPP) tracking was performed inside a custom-engineered nitrogen-filled enclosure, by using continuous one-sun illumination from a white LED array. The temperature was carefully maintained at around 25°C with the Multi-Channels Solar Cells Stability Test System from Wuhan 91PVKSolar Technology Co. Ltd, China. The system automatically recalculated the MPP at a certain time by continuously monitoring the current-voltage (J - V) characteristics.

Data Availability

The data generated in this study are provided in the Supplementary Information/Source Data file. All other data are available from the corresponding authors on request. Source data are provided with this paper.

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Author Contributions

Z.X. performed the device fabrication and the technology innovation. Y.D. conceived the idea and proposed the experimental design. Q.C., X.Z., J.Y., M.C., and Y.C. characterized the material samples (AFM, SEM, in-situ UV-vis and MPP test, etc.). J.Q. helped the fabrication of organic subcells. H.L. helped perform the GIWAXS measurements. Y.D. drafted the original manuscript. Y.D. and Q.P. supervised the project. All the authors discussed the results and commented on the manuscript.

Competing Interests Statement

The authors declare no conflict of interest.

Figure Legends

Fig. 1| The working mechanism of BiOCl on adsorbed Me-4PACz, and its influence on HSL.

a The adsorbed energy of Me-4PACz on BiOCl. **b,c** The in situ GIWAXS of **(b)** Me-4PACz and **(c)** Me-4PACz/BiOCl grown on Si substrate. **d** The schematic model of adsorbed Me-4PACz before and after BiOCl modification. **e,f** The KPFM images **(e)** and surface CPD distribution **(f)** of perovskite grown on Me-4PACz substrate before and after DMF rinsing and continuously heating at 85 °C for 200 h (Scale bar, 1 μ m). **g,h** The KPFM images **(g)** and surface CPD distribution **(h)** of perovskite grown on Me-4PACz/BiOCl substrate before and after DMF rinsing and continuously heating at 85 °C for 200 h (Scale bar, 1 μ m). **i,k** The CFM images of perovskite grown on **(i)** Me-4PACz and **(k)** Me-4PACz/BiOCl substrate before and after DMF rinsing and continuously heating at 85 °C for 200 h, respectively, (Scale bar in **i** and **k**, 1 μ m). **j,l** The surface current distribution.

Fig. 2| The influence and mechanism of BiOCl on perovskite crystallization. **a,b** The GIWAXS images of perovskite grown on **(a)** Me-4PACz and **(b)** Me-4PACz/BiOCl substrate. **c** The cross-sectional HAADF-STEM image of the BiOCl/perovskite interface (Scale bar, 2 nm). **d** Atomically resolved HAADF-STEM image of the BiOCl/perovskite interface (Scale bar, 500 pm). **e** Top and side views of the BiOCl(110)/perovskite(100) interface before and after the ion exchange, and the corresponding energy difference. **f** The XPS spectra of Pb 4*f* for PbI₂ and PbI₂/BiOCl films. **g** The TOF-SIMS curves of BiOCl-modified device. **h** The residual strain within perovskites. **i** FWHM of the steady-state PL spectra as a function of temperature for perovskite grown on Me-4PACz and Me-4PACz/BiOCl substrate.

Fig. 3| The device structure and photovoltaic performance of single-junction PSC. **a** The device structure of inverted PSC. **b** EQE_{EL} curves, **c** the fitting line of light intensity-dependent V_{oc} , and **d** $J-V$ curves with the inserted steady power output of control and BiOCl-modified 0.09 cm² devices. **e** The $J-V$ curves of control and BiOCl-modified 1.0 cm² devices. **f** The $J-V$ curves of BiOCl-modified devices with MeO-2PACz and 4PADCB as HSL. **g** The PCE tracking of encapsulated control and BiOCl-modified 0.09 cm² devices at MPP under the continuous illumination of the simulated light (AM1.5G, 100 mW cm⁻²) for 2400 h.

Fig.4| The device structure and photovoltaic performance of PO-TSC. **a** The device structure and **b** cross-sectional SEM images of PO-TSC. **c** $J-V$ curves of PSC, OSC and PO-TSC. **d** The steady-state power output of the champion PO-TSC. **e** The statistic PCE of TSC from 2022 to 2026. **f** The EQE curves of the perovskite and organic subcells in the champion PO-TSC. **g** Continuous MPP tracking of the encapsulated PO-TSC under simulated one-sun illumination (AM 1.5G,

100 mW cm⁻², without UV filter) in air atmosphere at room-temperature for 600 h.

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